Every Algebraic Number is a Critical Value of an Integer Polynomial

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Abstract

For every irreducible $f \in \mathbb{Z}[x]$ of positive degree we construct $g \in \mathbb{Z}[x]$ and an irreducible $F \in \mathbb{Z}[x]$ with $F \mid g'$ and $F^2 \mid f \circ g$. Equivalently, every algebraic number is a critical value of an integer polynomial, so the set of critical values of $\mathbb{Z}[x]$ is exactly $\overline{\mathbb{Q}}$. The construction is explicit. The proof was assisted by AI, and has been formalized in Lean.

1 Introduction

A critical value of a nonconstant $g \in \mathbb{Z}[x]$ is a number α with $g(\beta) = \alpha$ and $g'(\beta) = 0$ for some β. Since $\overline{\mathbb{Q}}$ is closed under polynomial evaluation, every critical value is algebraic. This note proves the converse, and does so with an explicit construction.

Theorem 1. *Let f be an irreducible integral polynomial of positive degree. Then there exist F and g in $\mathbb{Z}[x]$, with F irreducible and of positive degree, such that*

$$F \mid g' \quad \textit{and} \quad F^2 \mid f \circ g.$$

The corollary is the statement in the title.

Corollary 1. *The set of critical values is exactly $\overline{\mathbb{Q}}$.*

2 The construction

Let f be irreducible of degree $n \geq 1$ and α a root. Write $f(x) = \sum_{k=0}^{n} a_k x^{n-k}$, so that $a_n = f(0)$. If $f = x$, then $g = x^2$ and $F = x$ work. Otherwise $f(0) \neq 0$, and since $\gcd(f, f') = 1$ in $\mathbb{Q}[x]$, the elimination ideal $(f, f') \cap \mathbb{Z}$ is nonzero; let $u, v \in \mathbb{Z}[x]$ and $\rho \in \mathbb{Z}$, $\rho \neq 0$, be such that $u\, f + v\, f' = \rho$, and set $M := \rho\, a_n$, $F(x) := f(Mx)/a_n$ and $W(x) := (v(0)F(x) - v(Mx))/\rho$. The polynomial claimed by Theorem 1 is

$$g(x) := Mx + f(Mx)\, W(x). \tag{1}$$

Theorem (restatement of Theorem 1). *F is irreducible, and $F \mid g'$ and $F^2 \mid f \circ g$.*

pf. Write $F(x) = \sum_{k=0}^{n} A_k x^{n-k}$. From $A_k = a_k M^{n-k}/a_n$ we get $A_n = 1$ and $A_k = a_k \rho^{n-k} a_n^{n-k-1}$ for $k < n$, so $F \in \mathbb{Z}[x]$ with $F(0) = 1$, which gives $W(0) = 0$. And ρ divides M and every A_k with $k < n$, so it divides $v(0)F(x) - v(Mx)$ and $W \in \mathbb{Z}[x]$. Therefore g is an integral polynomial.

Let $\beta = \alpha/M$, a root of F; therefore $g(\beta) = \alpha$, and $\rho W(\beta) = -v(\alpha)$. Then

$$\begin{aligned} g'(\alpha/M) &= M + \rho\, a_n f'(\alpha)\, W(\alpha/M) + \cancel{f(\alpha)}\, W'(\alpha/M) \\ &= M - a_n f'(\alpha) v(\alpha) = 0, \end{aligned}$$

because $u(\alpha)\cancel{f(\alpha)} + v(\alpha) f'(\alpha) = \rho$. Hence β is a double root of $f \circ g$, and F is primitive irreducible, so $F \mid g'$ and $F^2 \mid f \circ g$ by Gauss. ■

Any ρ will do, but $M = \rho\, a_n$ multiplies into every coefficient of F, W and g, so the smallest gives the smallest answer: the generator of $(f, f') \cap \mathbb{Z}$, the reduced resultant, of which $\operatorname{Res}(f, f')$ is a multiple.

3 Examples

Each line below is a closed identity, checkable by expansion.

Example 1 ($f = qx - p$, $p \neq 0$, $\gcd(p, q) = 1$). Here $f' = q$ and $p = qx - f$, so $(f, f') \cap \mathbb{Z} = \mathbb{Z}$ and $\rho = 1$. Choose $a, b \in \mathbb{Z}$ with $ap + bq = 1$ and take $u = -a$, $v = ax + b$; then $f(0) = -p$, $M = -p$, $F = qx + 1$, $W = x$, and

$$g = -pqx^2 - 2px, \qquad g' = -2p\,(qx + 1), \qquad f \circ g = -p\,(qx + 1)^2.$$

Example 2 ($f = ax^2 + bx + c$). This is irreducible of degree 2 exactly when $a \neq 0$, $\gcd(a, b, c) = 1$, and $D := b^2 - 4ac$ is not a square. Since $(f')^2 = 4af + D$,

$$\rho = \operatorname{Res}(f, f') = -aD, \qquad u = 4a^2, \qquad v = -af',$$

so $M = -acD$ and

$$F = a^3cD^2x^2 - abD\,x + 1, \qquad W = a^3bcD\,x^2 - a(b^2 - 2ac)\,x,$$

$$g = a^6bc^3D^3x^4 - 2a^4c^2(b^2 - ac)D^2x^3 + a^2bc(b^2 - ac)D\,x^2 - 2ac(b^2 - 3ac)\,x,$$

with $g' = 2ac(2a^2bcD\,x - (b^2 - 3ac))F$. For $b = 0$ the quartic term drops and $\deg g = 3 = n + 1$, the least any solution can have.

Example 3 ($f = x^3 - x - 1$). The construction gives $M = 23$ and a sextic. A hand-tuned quintic also witnesses the *statement*, and is a useful independent check of it:

$$g = -24334x^5 + 39675x^4 - 26450x^3 + 9085x^2 - 1610x + 118, \qquad F = 23x^3 - 23x^2 + 8x - 1,$$

with $g' = -230\,(23x - 7)F$ and $F^2 \mid f \circ g$, the cofactor having degree 9.

Example 4 (Dedekind's cubic). $f = x^3 - x^2 - 2x - 8$, the standard example of a field whose ring of integers is not monogenic: 2 is a common index divisor [Ded78]. With $u = 21x - 125$, $v = -7x^2 + 44x - 3$, $\rho = 1006$, $f(0) = -8$, $M = -8048$,

$$F = 65158925824x^3 + 8096288x^2 - 2012x + 1, \qquad W = -194310912x^3 + 426544x^2 + 358x,$$

$$\begin{aligned} g = {}& 101288722414414331904x^6 - 209759614012620800x^5 \\ & - 217370176548864x^4 - 14767629312x^3 + 2350016x^2 - 10912x. \end{aligned}$$

4 Formalization

The development above has been formalized in Lean 4 [MU21] against Mathlib [The20], in about 1,200 lines. The main statement is

```
theorem main (f : Z[X]) (hf : 1 <= f.natDegree) :
    exists g H : Z[X], Irreducible H /\ H | derivative g /\ H^2 | f.comp g
```

with a strengthened form carrying $2 \leq \deg g$ and $1 \leq \deg H$, and with Corollary 1 proved from it. Every result depends only on the three standard axioms `propext`, `Classical.choice` and `Quot.sound`.

References

[Ded78] Richard Dedekind. “Über den Zusammenhang zwischen der Theorie der Ideale und der Theorie der höheren Kongruenzen”. In: *Abhandlungen der Königlichen Gesellschaft der Wissenschaften zu Göttingen* (1878).

[The20] The mathlib Community. “The Lean Mathematical Library”. In: *Certified Programs and Proofs (CPP 2020)*. ACM, 2020.

[MU21] Leonardo de Moura and Sebastian Ullrich. “The Lean 4 Theorem Prover and Programming Language”. In: *Automated Deduction – CADE 28*. Lecture Notes in Computer Science. Springer, 2021.